Detailed Math Behind Markov Chain Spectral Decomposition

1. The Setup: Transition Matrix and State Evolution

A Markov chain has a **transition matrix** P where entry P_{ij} is the probability of going from state j to state i. The state vector $\vec{p}(t)$ is a probability distribution — each entry $p_i(t)$ is the probability of being in state i at time t.

After one step:

$$\vec{p}(1) = P\,\vec{p}(0)$$

After two steps:

$$\vec{p}(2) = P\,\vec{p}(1) = P(P\,\vec{p}(0)) = P^2\,\vec{p}(0)$$

By induction, after t steps:

$$\vec{p}(t) = P^t\,\vec{p}(0)$$

The question is: how do we actually compute P^t and understand what it does? That's where eigendecomposition comes in.

2. Eigendecomposition of P

An eigenvector $\vec{v}_i$ of P satisfies:

$$P\,\vec{v}_i = \lambda_i\,\vec{v}_i$$

Multiplying by P just scales $\vec{v}_i$ by λ_i. So applying P repeatedly gives:

$$P^t\,\vec{v}_i = \lambda_i^t\,\vec{v}_i$$

This is easy to verify by induction: $P^2\vec{v}_i = P(P\vec{v}_i) = P(\lambda_i\vec{v}_i) = \lambda_i P\vec{v}_i = \lambda_i^2\vec{v}_i$, and so on.

Now suppose P has N linearly independent eigenvectors $\vec{v}_1, \dots, \vec{v}_N$. We can **decompose** the initial state in this eigenbasis:

$$\vec{p}(0) = c_1\,\vec{v}_1 + c_2\,\vec{v}_2 + \cdots + c_N\,\vec{v}_N$$

where the coefficients c_i are found using the **left eigenvectors** $\vec{u}_i$ (which satisfy $\vec{u}_i^T P = \lambda_i \vec{u}_i^T$). Because $\vec{u}_i^T \vec{v}_j = \delta_{ij}$ (they form a biorthogonal system), we get:

$$c_i = \vec{u}_i^T\,\vec{p}(0)$$

Now apply P^t:

$$\vec{p}(t) = P^t\,\vec{p}(0) = P^t \left(\sum_i c_i\,\vec{v}_i \right) = \sum_i c_i\,P^t \vec{v}_i = \sum_{i=1}^{N} \lambda_i^t\,c_i\,\vec{v}_i$$

This is the key formula. Each eigenvector contributes independently, and its amplitude changes by a factor of λ_i at every step.

3. Why $\lambda_1 = 1$ Is the Steady State

For a well-behaved (irreducible, aperiodic) Markov chain, the Perron–Frobenius theorem guarantees that the largest eigenvalue is $\lambda_1 = 1$, and all other eigenvalues satisfy $|\lambda_i| < 1$.

The corresponding eigenvector $\vec{v}_1 = \vec{\pi}$ is the **stationary distribution**: $P\vec{\pi} = \vec{\pi}$.

So as $t \to \infty$:

$$\vec{p}(t) = \underbrace{1^t \cdot c_1 \cdot \vec{\pi}}_{\text{survives}} + \underbrace{\lambda_2^t \cdot c_2 \cdot \vec{v}_2 + \cdots}_{\to 0}$$

Every term except the first decays to zero, because $|\lambda_i|^t \to 0$ for $|\lambda_i| < 1$. What remains is $c_1\vec{\pi}$, and since $\vec{p}(t)$ must sum to 1 and $\vec{\pi}$ sums to 1, we get $c_1 = 1$.

Therefore:

$$\lim_{t\to\infty} \vec{p}(t) = \vec{\pi}$$

regardless of the starting distribution. The system always converges to the stationary distribution.

4. How λ_i^t Gives Timescales

Rewrite the evolution as:

$$\vec{p}(t) = \vec{\pi} + \sum_{i=2}^{N} \lambda_i^t\,c_i\,\vec{v}_i$$

The "distance from equilibrium" is entirely captured by the sum on the right. Each mode i decays like λ_i^t.

Now rewrite λ_i^t using exponentials. Since $\lambda_i^t = e^{t \ln \lambda_i}$, we can define:

$$\tau_i = \frac{-1}{\ln(\lambda_i)}$$

so that:

$$\lambda_i^t = e^{-t/\tau_i}$$

This is exactly the form of exponential decay with **time constant** τ_i. At $t = \tau_i$, the amplitude has decayed by a factor of e:

$$\lambda_i^{\tau_i} = e^{-\tau_i/\tau_i} = e^{-1} \approx 0.368$$

So τ_i is how many steps it takes for mode i to lose about 63% of its signal.

Concrete example: if $\lambda_2 = 0.99$, then $\tau_2 = -1/\ln(0.99) \approx 99.5$ steps. That means the slowest non-trivial mode takes roughly 100 steps to decay significantly. If $\lambda_3 = 0.5$, then $\tau_3 = -1/\ln(0.5) \approx 1.44$ steps — it vanishes almost immediately.

5. The Spectral Gap and Metastability

The **spectral gap** is defined as $\gamma = 1 - \lambda_2$. A small gap (i.e., λ_2 close to 1) means the system mixes slowly.

Now consider what happens when there's a **large gap** between λ_2 and λ_3:

$$\lambda_2 \gg \lambda_3 \geq \lambda_4 \geq \cdots$$

At intermediate times where $\lambda_3^t \approx 0$ but λ_2^t is still significant, the state simplifies to:

$$\vec{p}(t) \approx \vec{\pi} + \lambda_2^t \, c_2 \, \vec{v}_2$$

The system effectively lives in a **two-dimensional subspace** spanned by $\vec{\pi}$ and $\vec{v}_2$. Since $\vec{v}_2$ must have both positive and negative entries (its entries sum to zero because it's orthogonal to the all-ones left eigenvector of λ_1), it naturally **partitions** the states into two groups: those where $v_{2,j} > 0$ and those where $v_{2,j} < 0$.

These two groups are the **metastable states**. The system stays trapped in one group for roughly τ_2 steps before transitioning. All fast modes decay quickly (small τ_i), so the internal structure within each group equilibrates fast, but the *slow* mode (λ_2) governs the rare transitions between groups.

More generally, if the first k eigenvalues are close to 1 and there's a gap before λ_{k+1}, then the chain has approximately k metastable clusters, and the dynamics at intermediate timescales live in a k-dimensional subspace.

6. Summary of the Full Picture

$$\underbrace{\vec{p}(t)}_{\text{state at time } t} = \underbrace{\vec{\pi}}_{\text{equilibrium}} + \underbrace{e^{-t/\tau_2}\, c_2\, \vec{v}_2}_{\text{slow mode (metastable)}} + \underbrace{\sum_{i\geq 3} e^{-t/\tau_i}\, c_i\, \vec{v}_i}_{\text{fast modes (transient)}}$$

The eigenvalues tell you *how fast* each pattern decays. The eigenvectors tell you *which pattern* decays. And gaps in the spectrum tell you where the system has natural bottlenecks or clusters.

Eigenvalue	**Meaning**	**Timescale**
$\lambda_1 = 1$	Steady state — the system has fully equilibrated	$\tau = \infty$
$\lambda_2 \approx 1$	Slowest mode — takes a long time to decay	$\tau_2 = -1/\ln(\lambda_2)$ (long)
$\lambda_i \ll 1$	Fast modes — decay quickly, transient dynamics	τ_i (short)